

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Artificial Intelligence

2. Subject Name: Deep Learning Foundations
Credits: 4

L-T-P: 3-1-0

3. Pre-requisites: Nil

Syllabus and reference books:

Syllabus:

Perceptron and Feedforward net	Theory	Introduction, Perceptron and Feed Forward Neural Network: Structure & Components, Activation function and non-linearity
	Tutorial	Introduction to pytorch
Optimization and Backpropagation	Theory	Gradient over graph, Optimization Algorithms for Deep Learning, Backpropagation Algorithm
	Tutorial	Working with gradient based optimizers, training deep nets with backpropagation
Normalization and Regularization	Theory	Initialization, Error functions, Regularization, Batch normalization
	Tutorial	Regularization and normalization
Convolutional Neural Nets	Theory	Building blocks of convolutional neural nets, Basic CNN, Deep CNN, VGG16, Residual networks
	Tutorial	Implementing and training basic CNN, VGG nets and Resnets
Recurrent Neural Nets	Theory	Recurrent neural net (RNN), LSTM, GRU, Backpropagation through Time
	Tutorial	Implementing and training RNN and LSTM
Attention Network	Theory	Encoder-decoder model and Attention nets
	Tutorial	Implementing attention nets
Self-Attention and Transformer	Theory	Transformers, BERT, Machine translation and Language models with transformers
	Tutorial	Implementing transformer and BERT, Machine translation with transformer
Pruning and Architecture Search	Theory	Neural net pruning (Structured and Unstructured approaches), Neural Architecture Search
	Tutorial	Neural architecture search
Advanced Learning with Deep Nets	Theory	Generative Adversarial learning, Multitask learning
	Tutorial	Implementing GAN, Multitask learning with

			CNN and Transformer	
Reference Books: 1. Deep Learning, Aaron Courville, Ian Goodfellow, Yoshua Bengio, MIT Press, 2016. 2. Linear Algebra and Learning from Data, Gilbert Strang, Wellesley Cambridge Press, 2019. 3. Neural Networks and Deep Learning: A Textbook, Charu C Aggarwal, Springer, 2018. 4. Optimization for Data Analysis, Wright and Recht, Cambridge University Press, 2022.				

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Perceptron and Feedforward net	5
2.	Optimization and Backpropagation	6
3.	Normalization and Regularization	3
4.	Convolutional Neural Nets	8
5.	Recurrent Neural Nets	5
6.	Attention Network	5
7.	Self-Attention and Transformer	7
8.	Pruning and Architecture Search	5
9.	Advanced Learning with Deep Nets	4
Total number of hours		39